Stieltjes Electrostatics and the Pandrosion Quotient: A Physical Dictionary for Polynomial Dynamics

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Abstract

We establish a complete dictionary between the Pandrosion quotient framework (Papers 1–25) and the classical Stieltjes electrostatic model of polynomial roots. The Pandrosion quotient $Q(\alpha_k, z) = P(z)/(z - \alpha_k)$ is the "weighted force" from the charge α_k; the total field is $F(z) = P'(z)/P(z) = \sum Q(\alpha_k, z)/P(z)$; the Pandrosion energy $E_P = \sum Q^2$ is the tidal field intensity times P^2; and Laguerre's inequality is the non-negativity of the tidal field. Most importantly, Newton's method is "moving along the total field" ($z' = z - 1/F$), while the Pandrosion iteration is "moving along a single-charge field" ($z' = z - 1/F_a$), explaining why Pandrosion is derivative-free.

1 The electrostatic model

Consider d unit point charges on the complex plane at positions $\alpha_1, \ldots, \alpha_d$ (the roots of P). The **Coulomb force** on a test particle at z from the charge α_k is:

$$F_k(z) = \frac{1}{z - \alpha_k}. \tag{1}$$

Proposition 1.1 (Pandrosion–Coulomb correspondence)**.**

$$F_k(z) = \frac{1}{z - \alpha_k} = \frac{Q(\alpha_k, z)}{P(z)}. \tag{2}$$

The Pandrosion quotient $Q(\alpha_k, z)$ is the ***weighted force****: the Coulomb force F_k multiplied by the total "potential amplitude" $P(z)$.*

2 The electrostatic dictionary

Theorem 2.1 (Complete Pandrosion–electrostatic dictionary)**.**

Electrostatics	*Pandrosion formula*	*Paper*
Force from α_k	$F_k = Q(\alpha_k, z)/P(z)$	*14*
Total field	$F = P'/P = \sum Q/P$	*14*
Equilibrium ($F = 0$)	$P'(\beta) = 0$	*14*
Influence weight	$\lambda_k = \lvert F_k\rvert^2 / \sum \lvert F_j\rvert^2$	*14*
Tidal field	$T = \sum F_k^2 = E_P/P^2$	*20*
$T \geq 0$ *(real line)*	$(P')^2 \geq PP''$	*20*
$T > 0$ *(simple roots)*	$E_P(x) > 0$	*23*
k*-body interaction*	$e_k(F) = P^{(k)}/(k!\,P)$	*21*
Self-energy at root	$E_P(\alpha_k) = P'(\alpha_k)^2$	*20*
Total self-energy	$\prod E_P(\alpha_k) = D(P)^2$	*22*
Spectral ratio	$r_k = F_k/F_a = Q(\alpha_k)/Q(a)$	*25*
Contraction	$C = \prod(1 - r_k)$	*25*

Proof. All identities follow from $F_k = Q(\alpha_k, z)/P(z) = 1/(z - \alpha_k)$ and the results of Papers 14, 20, 21, 22, 25. The tidal field identity:

$$T(z) = \sum_{k=1}^{d} F_k(z)^2 = \sum \frac{1}{(z - \alpha_k)^2} = \frac{E_P(z)}{P(z)^2}$$

follows from $E_P = \sum Q^2$ and $Q/P = F$. □

3 Newton vs Pandrosion: the physical picture

Theorem 3.1 (Root-finding as force-following)**.**

(i) ***Newton's method*** *is "following the total field":*

$$z_{Newton} = z - \frac{1}{F(z)} = z - \frac{P(z)}{P'(z)}. \tag{3}$$

The step displaces z by $-1/F$: the inverse of the total electric field.

(ii) ***Pandrosion's method*** *is "following a single-charge field":*

$$z_{Pandrosion} = z - \frac{1}{F_a(z)} = z - \frac{P(z)}{Q(a, z)}, \tag{4}$$

where $F_a(z) = Q(a, z)/P(z) \approx F_k(z)$ when $a \approx \alpha_k$ (anchor near root k).

Remark 3.2 (Why Pandrosion is derivative-free). Newton requires the total field $F = P'/P$, which involves *all* d forces—equivalent to computing P'. Pandrosion requires only $F_a = Q(a, z)/P(z)$, which involves a *single* difference quotient—no derivative. Physically:

Newton = d-body field computation   Pandrosion = 1-body tracking.

This is why Pandrosion can bypass McMullen's impossibility (Paper 8): it does not attempt to resolve the full field, only to follow one charge at a time.

4 Gauss–Lucas as electrostatic equilibrium

Corollary 4.1 (Gauss–Lucas is equilibrium containment). *$P'(\beta) = 0$ means $\sum F_k(\beta) = 0$: the forces at β are in perfect balance. By defining $\lambda_k = |F_k(\beta)|^2 / \sum |F_j(\beta)|^2 > 0$ with $\sum \lambda_k = 1$, one shows:*

$$\beta = \sum_{k=1}^{d} \lambda_k \alpha_k \quad \in \quad \mathrm{conv}\{\alpha_1, \ldots, \alpha_d\}.$$

Physical interpretation: *At equilibrium, the test particle is at the "centre of gravity" of the charges (weighted by force squared). Equilibrium cannot occur outside the convex hull.*

5 Laguerre's inequality as tidal positivity

Corollary 5.1 (Laguerre = non-negative tidal field). *For real charges on the real line:*

$$T(x) = \sum_{k=1}^{d} \frac{1}{(x - \alpha_k)^2} \geq 0 \qquad \iff \qquad (P'(x))^2 \geq P(x)\, P''(x).$$

Physical interpretation: *The tidal field (stretching rate due to the charge configuration) is always non-negative on the real line. One-dimensional charges can stretch but never compress a test interval.*

6 The contraction ratio as field competition

In the electrostatic picture, the spectral ratios $r_k = F_k/F_a$ (Paper 25) measure the field of each charge *relative* to the anchor field. The contraction ratio $C = \prod(1 - r_k)$ represents the "resolution" of the step: when the anchor's field F_a dominates all others ($|r_k| < 1$), the step is well-resolved and $|\arg C| < \pi$.

The energy dominance condition $|Q(a, z)|^2 > E_P(z)$ (Paper 25) translates to:

$$|F_a|^2 > T(z) = \sum F_k^2 : \qquad \text{"the anchor is stronger than the tidal field."} \tag{5}$$

7 Conclusion

Every result of the Pandrosion series admits a physical interpretation:

- **Papers 14–18** (geometry): properties of equilibria in charge configurations.
- **Papers 19, 24** (stability): charge configurations relative to boundaries (disk, half-plane).
- **Papers 20–23** (energy): the tidal field and its global properties.
- **Papers 7–13, 25** (dynamics): root-finding as single-charge tracking in the electrostatic field.

The Pandrosion quotient $Q(\alpha_k, z) = P(z)/(z - \alpha_k)$ is the *weighted Coulomb force*. The entire theory of polynomial zeros—from Gauss–Lucas to Smale—is electrostatics.

References

[1] T. J. Stieltjes, *Sur certains polynômes qui vérifient une équation différentielle*, Acta Math. **6** (1885), 321–326.

[2] I. Besevic, *Derivative-free Gauss–Lucas via Pandrosion influence weights*, Preprint (2026).

[3] I. Besevic, *Laguerre's inequality from Pandrosion energy*, Preprint (2026).

[4] I. Besevic, *Argument control via energy dominance*, Preprint (2026).